

#1. Prove that $\tilde{L}^t(\cdot | \pi^t)$ is an "unbiased estimate" for all $\pi^t \in \Pi$

↳ $\mathbb{E}_{\pi^t \sim \Pi} [\tilde{L}^t(\cdot | \pi^t)] = L^t$

• Expected Return

- Expected Regret:

$$f[\text{reg}] = \mathbb{E} \left[\sum_{t=1}^T \langle p^t, \ell^t \rangle \right] - \min_{\pi \in [A]} \sum_{t=1}^T \langle e_{\pi}, \ell^t \rangle$$

$\text{LH9.}^* \quad \langle p^t, \ell^t \rangle = \sum_{\pi=1}^A p_t(\pi) \ell_t(\pi)$
 $e_{\pi}(j) = \begin{cases} 1 & j=\pi \\ 0 & j \neq \pi \end{cases}$

the optimal

- the estimator:

$$\tilde{\rho}^{\pm}(\pi | \pi_{\pm}) = \frac{\rho^{\pm}(\pi)}{\rho^{\pm}(\pi)} \mathbb{1}_{\{\pi = \pi_{\pm}\}}, \pi \in [A]$$

Part 1

$$\mathbb{E}_{\pi^{\pm} \pi^{\pm}} [\hat{\chi}^{\pm}(\pi | \pi^{\pm})] = \sum_{\pi' \in [A]} P^{\pm}(\pi') \cdot \tilde{\chi}^{\pm}(\pi | \pi') \quad (\text{또의 정리})$$

$$= \sum_{\pi' \in [A]} \underbrace{p^*(\pi')}_{\text{실제 분포}} \frac{l^*(\pi)}{\underbrace{p^*(\pi)}_{\text{실제 분포}}} \mathbb{1}_{\{\pi' = \pi\}} \quad (\text{estimator의 정의})$$

$$= p^\pm(\pi) \cdot \frac{l^\pm(\pi)}{p^\pm(\pi)}$$

$$= l^{\pm}(\pi)$$

$$\mathbb{E}_{\pi \sim p_t} [\tilde{\ell}^{\pm}(\cdot | \pi^t)] = \ell^{\pm}$$

Part 2

$$\mathbb{E}_{\pi^+ \sim p^+} [\langle p^+, x^+ \rangle] = \mathbb{E}_{\pi^+ \sim p^+} \left[\sum_{\pi} p^+(\pi) \hat{\ell}_+(\pi | \pi^+) \right] \quad (\text{바깥의 정의})$$

$$= \sum_{\pi} p^t(\pi) \sum_{\pi^t \sim p^t} \underbrace{\mathbb{E} \left[\mathbf{1}^t(w | \pi^t) \right]}_{\text{(Linearity of Expectation)}}$$

$$= \sum_{\pi} p^L(\pi) \underbrace{\ell^L(\pi)}_{\text{(Part 1)}}$$

$$= \langle p^t, l^t \rangle \text{ (내적의 정의)}$$

$$\therefore \frac{\hbar}{m \lambda_{\text{de Broglie}}} [\langle p^x, \hat{x}^x \rangle] = \langle p^x, \hat{x}^x \rangle$$

part 3

$$\begin{aligned} \langle e_{\pi}, \tilde{l}^t \rangle &= \sum_{j=1}^n e_{\pi(j)} \tilde{l}^t(j|\pi^t) \\ &= \underbrace{e_{\pi(j)} \tilde{l}^t(j|\pi^t)}_{\substack{\uparrow \\ 1}} + \sum_{j \neq \pi} \underbrace{e_{\pi(j)} \tilde{l}^t(j|\pi^t)}_{\substack{\uparrow \\ 0}} \\ &= \tilde{l}^t(\pi|\pi^t) \end{aligned}$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \langle p^t, \tilde{l}^t \rangle dt = \langle p^t, l^t \rangle$$

$$\#_{\pi^t \sim p^t} [\langle e_{\pi}, \tilde{l}^t \rangle] = \langle e_{\pi}, l^t \rangle \text{ (same with part 2)}$$

$$\sum_{t=1}^T \langle e_{\pi}, l^t \rangle = \sum_{t=1}^T \mathbb{E}_{l^t \sim p_t} [\langle e_{\pi}, \tilde{l}^t \rangle]$$

min & expectation.

$$\min_{\pi} \sum_{t=1}^T \langle \mathbf{p}_t, \mathbf{x}^t \rangle = \min_{\pi} \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi_t} [\langle \mathbf{p}_t, \mathbf{x}^t \rangle] \right]$$

($\mathbf{p}_t$ 는 learner의 random choice와
상관 $\mathbf{x}$, 상수임.)
(상수의 기댓값: 상수.)

Part 4

$$\mathbb{E}[\text{reg}] = \mathbb{E}\left[\sum_{t=1}^T \underbrace{\langle p_t, l_t \rangle}\right] - \min_{p \in [A]} \sum_{t=1}^T \underbrace{\langle p, l_t \rangle}.$$

$$= \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi_t^{\text{opt}}} [\langle p_t^*, \tilde{x}^t \rangle] \right] - \min_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi_t} [\langle e_t, \tilde{x}^t \rangle] \right]$$

now that given π ,

$$\mathbb{E}_{\pi \sim p^t} [\hat{\ell}(\pi | \pi')^2] = \frac{\ell^t(\pi')^2}{p^t(\pi')}, \text{ so that } \mathbb{E}_{\pi^t \sim p^t} \mathbb{E}_{\pi \sim p^t} [\hat{\ell}^t(\pi | \pi')^2] \leq A.$$

$$* \hat{\ell}^t(\pi | \pi') = \frac{\ell^t(\pi)}{p^t(\pi)} \mathbb{1}_{\{\pi' = \pi\}}.$$

Part 1.

$$\begin{aligned} \mathbb{E}_{\pi \sim p^t} [\hat{\ell}(\pi | \pi')^2] &= \mathbb{E}_{\pi \sim p^t} \left[\left(\frac{\ell^t(\pi)}{p^t(\pi)} \mathbb{1}_{\{\pi' = \pi\}} \right)^2 \right] \\ &= \sum_{\pi \in [A]} p^t(\pi) \left(\frac{\ell^t(\pi)}{p^t(\pi)} \mathbb{1}_{\{\pi' = \pi\}} \right)^2 \\ &= \sum_{\pi} p^t(\pi) \frac{\ell^t(\pi)^2}{p^t(\pi)^2} \mathbb{1}_{\{\pi' = \pi\}} \\ &= p^t(\pi') \cdot \frac{\ell^t(\pi')^2}{p^t(\pi')^2} \\ &= \frac{\ell^t(\pi')^2}{p^t(\pi')} \end{aligned}$$

Part 2.

$$\mathbb{E}_{\pi \sim p^t} [\hat{\ell}^t(\pi | \pi^t)^2] = \frac{\ell^t(\pi^t)^2}{p^t(\pi^t)} \quad (\text{from part 1})$$

$$\begin{aligned} \mathbb{E}_{\pi^t \sim p^t} \mathbb{E}_{\pi \sim p^t} [\hat{\ell}^t(\pi | \pi^t)^2] &= \mathbb{E}_{\pi^t \sim p^t} \left[\frac{\ell^t(\pi^t)^2}{p^t(\pi^t)} \right] \\ &= \sum_{\pi^t \in [A]} p^t(\pi^t) \frac{\ell^t(\pi^t)^2}{p^t(\pi^t)} \\ &= \sum_{\pi^t \in [A]} \ell^t(\pi^t)^2 \leq \sum_{\pi^t \in [A]} 1 = A. \\ &\quad \uparrow \\ &\quad \ell^t(\pi^t) \in [0, 1]. \end{aligned}$$

$$\therefore \mathbb{E}_{\pi^t \sim p^t} \mathbb{E}_{\pi \sim p^t} [\hat{\ell}^t(\pi | \pi^t)^2] \leq A.$$

Define $p^t(\pi) \propto \exp \left\{ -\eta \sum_{s=1}^t \langle e_\pi, \tilde{\ell}^s(\cdot | \pi^s) \rangle \right\}$.

Show that for any $\pi \in [A]$,
 estimated losses. $= \exp \left(-\eta \sum_{s=1}^t \tilde{\ell}^s(\pi | \pi^s) \right), \pi \in [A]$.

$$\mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} \langle p^t, \tilde{\ell}^t \rangle \right] - \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} \langle e_\pi, \tilde{\ell}^t \rangle \right] \leq \sqrt{AT \log A}.$$

↳ use (1.41):

$$\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} \left[\underbrace{\ell(\hat{f}^t(\pi^t), y^t)}_{\text{fixed comparator. (fixed action } \pi)} - \underbrace{\ell(\hat{f}^t(\pi), y^t)}_{\tilde{\ell}^t(\pi | \pi^t)} \right] \leq \frac{\eta}{2} \sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} [\ell(\hat{f}^t(\pi^t), y^t)^2] + \frac{\log |F|}{\eta}.$$

• loss of exponential weights algorithm (at time t):

$$\begin{aligned} \log_t &= \sum_{\pi} p^t(\pi) \ell^t(\pi) \\ &= \sum_{\pi} p^t(\pi) \frac{\ell^t(\pi)}{p^t(\pi)} \\ &= \sum_{\pi \in [A]} p^t(\pi) \frac{\ell^t(\pi)}{p^t(\pi)} \mathbb{1}_{\{\pi_t = \pi\}} \\ &= \sum_{\pi \in [A]} p^t(\pi) \tilde{\ell}^t(\pi | \pi^t) \\ &= \mathbb{E}_{\pi^t \sim p^t} [\tilde{\ell}^t(\pi^t | \pi^t)] = \langle p^t, \tilde{\ell}^t \rangle. \end{aligned}$$

• loss of the fixed comparator:
 $(p^t \text{ on } e_\pi)$

$$\ell_{\text{comp}, t} = \langle e_\pi, \tilde{\ell}^t \rangle = \tilde{\ell}^t(\pi | \pi^t)$$

(1.41) becomes...

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} [\tilde{\ell}^t(\pi^t | \pi^t)] - \sum_{t=1}^T \tilde{\ell}^t(\pi | \pi^t) &\leq \frac{\eta}{2} \sum_{t=1}^T \underbrace{\mathbb{E}_{\pi^t \sim p^t} [\tilde{\ell}^t(\pi^t | \pi^t)^2]}_{\substack{\downarrow \\ \langle p^t, \tilde{\ell}^t \rangle}} + \frac{\log A}{\eta} \leq \underbrace{\frac{\eta AT}{2} + \frac{\log A}{\eta}}_{\substack{\text{set } \eta = c \sqrt{\frac{\log A}{AT}} \\ \downarrow \\ \text{RHS} : \sqrt{AT \log A}}} \\ &= \mathbb{E}_{\pi^t \sim p^t} \mathbb{E}_{\pi^t \sim p^t} [\tilde{\ell}^t(\pi^t | \pi^t)^2] \leq A \quad (\text{from prob. 2}) \end{aligned}$$

take expectation.
 $\downarrow$

$$\mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} \langle p^t, \tilde{\ell}^t \rangle \right] - \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{\pi^t \sim p^t} \langle e_\pi, \tilde{\ell}^t \rangle \right] \leq \sqrt{AT \log A}$$